

Learning Plankton Community States with Restricted Boltzmann Machines

Francesco Pazzocco, Leonardo Pieripolli and Mattia Ponchio

University of Padova, Physics of Data, 35131 Padova, Italy

email: {francesco.pazzocco, leonardo.pieripolli,
mattia.ponchio}@studenti.unipd.it

Abstract—Automated underwater imaging now records lake plankton daily, but turning dozens of taxon counts into a small set of interpretable community states remains an open step, and the counts are overdispersed, zero-heavy and partly missing. Restricted Boltzmann machines (RBMs) offer a generative latent model with exact conditionals, yet have been applied to ecological assemblages almost only with binary presence-absence units. We train RBMs with Negative Binomial (NB) and zero-inflated NB visible units on 1906 clean days of 83 taxa from Lake Greifen (2019–2024), compare four hidden-unit types over 10 seeds per configuration, stabilise training with a log-mean clamp and persistent contrastive divergence, and validate the latent states against an independent archetypal analysis ($k = 5$). A one-hot (Softmax) hidden layer offered up to 10 states uses 6.2 ± 0.6 of them, and 6.5 ± 1.0 with zero-inflated visibles; its day labels agree with the dominant archetype with normalised mutual information 0.45 and 71% purity, and three archetypes map to a single state for 94–99% of their days. An NB RBM with 6 Sigmoid hidden units matches four of the five archetypes by cosine similarity (0.54–0.78) and is first or tied in masked-taxon reconstruction among the count families tested (mean absolute error 0.509 ± 0.017 on the unmasked taxa), while the Softmax layer pays about 30% in reconstruction for its interpretability. Count-likelihood RBMs therefore recover plankton community states that an independent method also finds, with roughly six states as the resolution supported by these data.

Index Terms—Restricted Boltzmann machine, Negative Binomial, plankton community states, persistent contrastive divergence, energy-based models

I. INTRODUCTION

Plankton communities respond within days to temperature, nutrients and grazing, and their composition is a standard indicator of lake water quality. Automated underwater microscopes now classify plankton images continuously: at Lake Greifen (Switzerland) a dual-magnification camera has produced daily abundances for 83 taxa over more than five years [1]. Ecologists summarise such data with a few recurrent *community states*, such as a spring diatom bloom or a summer assemblage of green algae and cyanobacteria. A model that finds these states directly from the counts, and that can also fill in taxa missing from a day, would turn the raw stream into something a limnologist can read and act on.

Two families of tools address this. Descriptive decompositions, such as archetypal analysis [2], express each day as a mixture of a few extreme communities and give interpretable states, but they define no probability model and cannot impute missing taxa. Generative latent-variable models learn $p(\mathbf{v})$ and

therefore handle both tasks. Among them, restricted Boltzmann machines (RBMs) [3] have exact factorised conditionals and few parameters, and each hidden unit can be read from its weights as a feature of the data. Negative Binomial (NB) likelihoods, possibly with zero inflation [4] (ZINB), are the standard description of overdispersed, zero-rich counts.

What has not been done is to combine the two for plankton time series. RBMs applied to species assemblages use binary presence-absence visible units [5], which discard abundance, the quantity that separates a bloom from background. The exponential-family extension of RBMs [6] allows count visibles, but NB and ZINB RBMs have, to our knowledge, not been evaluated on ecological time series, and the latent states such models learn have not been checked against an independent description of the same community. Raw plankton counts also make RBM training fragile: abundances span three orders of magnitude, and bloom and non-bloom days form well-separated modes that short Gibbs chains do not cross.

We train RBMs on the Lake Greifen series (1906 clean days, 83 taxa, 2019–2024) with binary, NB and ZINB visible units and four hidden-unit types (Bernoulli, Sigmoid read-out, Softmax, rectified Gaussian), always over 10 random seeds. We report the numerical fixes that training on raw counts requires, select the number of hidden units L on chronological and shuffled splits, measure how many states a one-hot hidden layer actually uses, compare the hidden units with an archetypal analysis ($k = 5$) computed independently on the same data, and evaluate reconstruction of masked taxa and of days with structurally missing taxa.

The contributions are:

- 1) We show that a one-hot (Softmax) hidden layer offered up to 10 states settles on about six, for both NB and ZINB visible units, and that its day-level labels agree with the dominant archetype (NMI 0.45, purity 71%), with three of five archetypes captured by a single state (Section IV-D, Section IV-E).
- 2) We show that an NB RBM with $L = 6$ recovers four of the five archetypes by cosine similarity and is first or tied in masked-taxon reconstruction among the six count families evaluated, while the Softmax layer costs about 30% in reconstruction (Section IV-E, Section IV-F).
- 3) We identify the training failures specific to raw counts (overflow of the NB mean, mode-locked CD-1 chains, unbounded ReLU units) and the fixes that make every

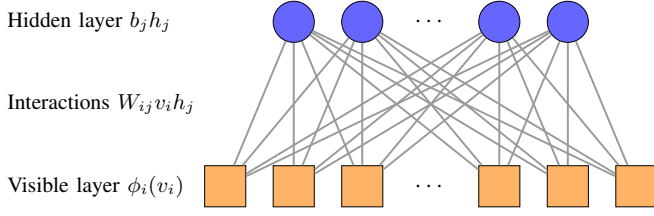


Fig. 1. An RBM is bipartite: every visible unit v_i (squares, one per taxon) is coupled to every hidden unit h_j (circles) and there are no couplings within a layer. Labels give the corresponding terms of the energy (1).

bounded family train in 10/10 seeds up to $L = 10$ (Section IV-B).

- 4) We evaluate imputation on 160 days whose missing taxa follow instrument retraining blocks, and show where count RBMs fail: bloom-forming taxa dominate the reconstruction error (Section IV-F).

Section II recalls RBMs, Gibbs sampling and persistent contrastive divergence, Section III describes data, models and protocol, Section IV reports the results, and Section V discusses limits and next steps.

II. THEORY

A. Restricted Boltzmann Machines

A restricted Boltzmann machine [3] is a Markov random field on two layers: D visible units $\mathbf{v}$, one per observed variable (here one per taxon), and L hidden units $\mathbf{h}$ that carry the latent representation. Units are coupled only across layers (Fig. 1), and a configuration has energy

$$E(\mathbf{v}, \mathbf{h}) = - \sum_i \phi_i(v_i) - \sum_j b_j h_j - \sum_{i,j} v_i W_{ij} h_j, \quad (1)$$

where ϕ_i contains the visible bias (for binary units $\phi_i(v_i) = a_i v_i$), b_j are the hidden biases and W_{ij} the couplings. The model assigns $p(\mathbf{v}, \mathbf{h}) = e^{-E(\mathbf{v}, \mathbf{h})}/Z$, with partition function $Z = \sum_{\mathbf{v}, \mathbf{h}} e^{-E(\mathbf{v}, \mathbf{h})}$ (an integral for continuous units), and the distribution of the data is the marginal $p(\mathbf{v}) = \sum_{\mathbf{h}} p(\mathbf{v}, \mathbf{h}) = e^{-F(\mathbf{v})}/Z$, where F is the free energy. For binary hidden units the sum over $\mathbf{h}$ factorises and

$$F(\mathbf{v}) = - \sum_i \phi_i(v_i) - \sum_j \text{softplus}\left(b_j + \sum_i W_{ij} v_i\right), \quad (2)$$

so $p(\mathbf{v})$ is known up to Z , which for $D = 83$ count variables cannot be computed.

The absence of intra-layer couplings is what makes the model usable: given one layer, the units of the other are conditionally independent,

$$p(\mathbf{h} | \mathbf{v}) = \prod_j p(h_j | \mathbf{v}), \quad p(\mathbf{v} | \mathbf{h}) = \prod_i p(v_i | \mathbf{h}), \quad (3)$$

with, for binary units, $p(h_j=1 | \mathbf{v}) = \sigma(b_j + \sum_i W_{ij} v_i)$ and $p(v_i=1 | \mathbf{h}) = \sigma(a_i + \sum_j W_{ij} h_j)$. Each hidden unit therefore acts as a detector of one pattern of co-varying taxa, and a day is described by which detectors fire; with few hidden units each feature can be read directly from its weights.

B. Likelihood Gradient

Maximum likelihood on training days $\{\mathbf{v}^{(n)}\}$ maximises $\langle \log p(\mathbf{v}) \rangle_{\text{data}}$. For any parameter ϑ ,

$$\frac{\partial \langle \log p(\mathbf{v}) \rangle_{\text{data}}}{\partial \vartheta} = - \left\langle \frac{\partial E}{\partial \vartheta} \right\rangle_{\text{data}} + \left\langle \frac{\partial E}{\partial \vartheta} \right\rangle_{\text{model}}, \quad (4)$$

where the first average uses the data with $\mathbf{h} \sim p(\mathbf{h} | \mathbf{v})$ (positive phase) and the second the model distribution $p(\mathbf{v}, \mathbf{h})$ (negative phase). For a coupling, $\partial E / \partial W_{ij} = -v_i h_j$, so learning matches data and model correlations. The positive phase is exact thanks to Eq. (3); the negative phase is an expectation under $p(\mathbf{v}, \mathbf{h})$, as intractable as Z , and must be estimated by sampling [7].

C. Gibbs Sampling as a Markov Chain

Equation (3) gives an exact block Gibbs sampler [8]: from a state $\mathbf{v}^{(t)}$, draw $\mathbf{h}^{(t)} \sim p(\mathbf{h} | \mathbf{v}^{(t)})$ and then $\mathbf{v}^{(t+1)} \sim p(\mathbf{v} | \mathbf{h}^{(t)})$, each layer in one parallel step. The sequence $\mathbf{v}^{(0)}, \mathbf{v}^{(1)}, \dots$ is a Markov chain with transition kernel

$$T(\mathbf{v}' | \mathbf{v}) = \sum_{\mathbf{h}} p(\mathbf{v}' | \mathbf{h}) p(\mathbf{h} | \mathbf{v}). \quad (5)$$

Each half-step leaves $p(\mathbf{v}, \mathbf{h})$ invariant, so $p(\mathbf{v})$ is a stationary distribution of T ; since all conditionals are strictly positive the chain is irreducible and aperiodic, hence ergodic, and the distribution of $\mathbf{v}^{(t)}$ converges to $p(\mathbf{v})$ for any starting point. What the theory does not fix is the speed. The distance from stationarity decays roughly as $e^{-t/\tau}$, with a mixing time τ set by the spectral gap of T , and τ grows sharply when $p(\mathbf{v})$ has well-separated modes: moving between them requires passing through low-probability configurations, which a local sampler rarely does. Plankton data are a case in point: bloom days and background days occupy distant regions of count space, and as the model learns to represent both, the energy barrier between them grows.

D. Contrastive Divergence and Persistent Chains

Running a chain to equilibrium at every parameter update is too slow, so the negative phase is approximated. Contrastive divergence (CD- k) [3] starts the chain at the current training batch and takes k Gibbs steps, usually $k = 1$. The resulting gradient is biased: if $k \ll \tau$ the samples stay near the data, and in particular a chain started from a bloom day remains in the bloom mode. The negative phase then never sees the configurations the model wrongly favours far from the data, and nothing pushes their probability down. The bias grows with the separation of the modes, which is why CD-1 becomes unreliable as L , and with it the model's ability to represent separate modes, increases (Section IV-B).

Persistent contrastive divergence (PCD) [9] keeps a set of chains alive across updates: at each step the chains continue from where they stopped, under the new parameters. This is a stochastic-approximation scheme: if the parameters change slowly compared with the mixing of the chains, the persistent chains remain close to the current model distribution, and their state accumulates many Gibbs steps over training. They can

therefore cross between modes that a one-step chain started at the data never reaches. The approximation degrades when the learning rate is large or the mixing is extremely slow, and the training can then stay out of equilibrium in ways that shape the learned model. We use PCD with 500 chains and a decaying learning rate (Section III-C).

E. Count Visible Units

The energy (1) admits any visible base measure ϕ_i , which yields the exponential-family harmoniums of [6]. In such a harmonium, $p(v_i | \mathbf{h})$ belongs to an exponential family whose natural parameter is the linear input $a_i + \sum_j W_{ij}h_j$. For a Negative Binomial with dispersion θ_i , the natural parameter is $\log(\mu_i/(\mu_i + \theta_i))$, which must be negative.

We instead adopt the parametrisation standard for overdispersed counts in regression: the NB mean has a log link, $\mu_i = \exp(a_i + \sum_j W_{ij}h_j)$ (Eq. (6)), so the input sets the log-mean directly and is unconstrained. The price is that this pair of conditionals, NB with log link and logistic hidden units, does not derive from a single energy of the form (1). The model is a conditional specification trained with the CD/PCD scheme of Eq. (4), in which the visible statistic v_i is replaced by the score of the NB log-likelihood with respect to the log-mean (Eq. (7)). The conditional factorisation and the block Gibbs sampler are unchanged, which is what sampling, imputation and reconstruction require; the chain is still ergodic and has a unique stationary distribution, but that distribution is defined by the sampler rather than by an energy; the free energy (2) and pseudo-likelihoods computed from it are exact only for the binary baseline. Zero inflation [4] adds a structural-zero probability π_i on top of the NB.

III. METHODS

A. Dataset and Preprocessing

We use the daily plankton abundances of the Lake Greifen imaging station [1]: 2113 days from 2019-03-21 to 2024-12-31, 83 taxa, in organisms per μL (range $[0, 0.444]$, 25.7% zeros). Total daily abundance is dominated by biology rather than instrument drift (coefficient of variation 1.29, dominant Lomb–Scargle period ≈ 365 days, stable annual median); an episode of $\approx 7\times$ the mean abundance in December 2022–February 2023 is kept as a real winter bloom, since the data descriptor reports no instrument issue in that window [1].

Four preprocessing steps are applied. (i) The 47 all-zero days are dropped. (ii) The 160 days with missing taxa are set aside as a separate test set: their gaps are not random but follow classifier-retraining blocks, in three patterns with 3, 31 and 54 missing taxa (104, 43 and 13 days). (iii) Rows are not normalised, so total biomass remains a signal. (iv) Values are multiplied by $s = 1000$, which brings them to an approximately integer scale (range $[0, 444]$); at sub-unit values the gradient of the NB log-likelihood with respect to the dispersion vanishes and the dispersion never leaves its initial value. The resulting 1906 clean days are split 85/15 into training and validation in two ways. The *chronological* split validates on the last 15% of days (2024), an honest test

of generalisation to an unseen year. The *shuffled* split draws validation days at random, with the run’s seed, so that every season appears in training; we use it to discover states, and the chronological split to select L conservatively.

For the binary baseline each taxon is binarised at its training-set median, which gives balanced units; a zero threshold makes common taxa almost always active and the task trivial.

B. Models

All models share the bipartite structure of Section II-A, with $D = 83$ visible units and L hidden units, and differ in the conditional distribution of each layer.

Visible units. The binary baseline (BB-RBM) has $p(v_i=1 | \mathbf{h}) = \sigma(a_i + \sum_j W_{ij}h_j)$. The NB visible units have

$$p(v_i | \mathbf{h}) = \text{NB}(v_i; \mu_i(\mathbf{h}), \theta_i), \quad \mu_i(\mathbf{h}) = e^{\eta_i}, \quad (6)$$

with log-mean $\eta_i = a_i + \sum_j W_{ij}h_j$, mean μ_i , variance $\mu_i + \mu_i^2/\theta_i$ and a learned dispersion $\theta_i > 0$ per taxon; the log-gamma form of the NB mass function accepts the non-integer scaled counts. The ZINB units add a structural-zero probability π_i : $p(v_i | \mathbf{h}) = \pi_i \delta_0(v_i) + (1 - \pi_i) \text{NB}(v_i; \mu_i(\mathbf{h}), \theta_i)$.

Hidden units. Given $\mathbf{v}$, the hidden input is $x_j = b_j + \sum_i W_{ij}v_i$. We compare four types. *Bernoulli*: $p(h_j=1 | \mathbf{v}) = \sigma(x_j)$, sampled as binary. *Sigmoid*: the same Bernoulli units, trained identically, but every deterministic read-out (reconstruction, hidden representation) uses the activation probability $\sigma(x_j)$ instead of a sample, a mean-field read-out. *Softmax*: $p(\mathbf{h} = \mathbf{e}_j | \mathbf{v}) = \text{softmax}_j(\mathbf{x})$, exactly one unit active per day, which turns the RBM into a mixture model with one state per day. *ReLU*: rectified Gaussian units [10], $h_j = \min(\max(0, x_j + \epsilon), 5)$, $\epsilon \sim \mathcal{N}(0, 1)$. Crossing visible and hidden types gives the ten families of Tab. 2.

C. Training

The likelihood gradient is the difference between a data-driven positive phase and a model-driven negative phase. For NB visibles, the score with respect to η_i is the weighted residual $r_i = \theta_i(v_i - \mu_i)/(\mu_i + \theta_i)$, so

$$\Delta W_{ij} \propto \langle h_j r_i \rangle_+ - \langle h_j r_i \rangle_-, \quad (7)$$

and analogously for $\mathbf{a}$ and $\mathbf{b}$; θ_i (and π_i) are updated by automatic differentiation of the positive-phase NLL with a learning rate ten times smaller. The negative phase uses PCD-1 [9] with 500 persistent chains initialised from training days; at each mini-batch a random subset of chains takes one Gibbs step and is written back.

All families share one protocol (Tab. 1): RMSprop ($\beta = 0.9$) with initial learning rate 0.01 decayed by 0.998 per epoch, 20 mini-batches per epoch with a batch size growing quadratically from 10 to 256, ℓ_1 penalty 10^{-4} on W , and 500 epochs (1000 for ZINB-Sigmoid and ZINB-Softmax). Two guards keep training finite: η_i is clamped at 10 before the exponential, so $\mu_i \leq 22\,026$, 50 times the data maximum, which never binds on real data; and ReLU activations are clamped to $[0, 5]$. Every (family, L , split) is trained with 10

TABLE 1. Training protocol, identical for all families unless stated.

Parameter	Value
Optimiser	RMSprop, $\beta = 0.9$, $\epsilon = 10^{-4}$
Learning rate	$0.01 \times 0.998^{\text{epoch}} (\theta, \pi: \times 0.1)$
Mini-batches	20/epoch, size 10 $\rightarrow$ 256 (quadratic)
Negative phase	PCD-1, 500 persistent chains
Regularisation	$\ell_1 = 10^{-4}$ on W
Epochs	500 (ZINB-Sigmoid/Softmax: 1000)
Guards	$\eta \leq 10$; ReLU $h \in [0, 5]$; $\theta \geq 10^{-4}$
Seeds	10 per (family, L , split)
Count scale	$s = 1000$

seeds; the spread across seeds therefore reflects initialisation and sampling only for the chronological split, and additionally the validation draw for the shuffled one.

D. Evaluation Protocol and Metrics

Fit. For count models we monitor the validation NLL per entry, $-\frac{1}{ND} \sum_{n,i} \log p(v_i^{(n)} | \mathbf{h}^{(n)})$ with $\mathbf{h}^{(n)} \sim p(\mathbf{h} | \mathbf{v}^{(n)})$, a one-step reconstruction likelihood that needs no partition function; lower is better. It is not normalised: through the $\ln \Gamma(v_i + 1)$ term it depends on the count scale s and on the data, so it has no absolute meaning. We use it only to check convergence and to compare models trained on the same data and scale (choice of L , Tab. 2); the substantive comparisons rest on reconstruction error and on the archetype analysis. Binary models are monitored with the negative pseudo-log-likelihood [11], which is not on the same scale. L is increased only if the gain exceeds one standard deviation across seeds.

States. For Softmax layers a unit is *used* if its mean activation over the training days exceeds 0.05 (the uniform share at $L = 10$ is 0.1). For Sigmoid read-outs a unit is *on* if $\sigma(x_j) \geq 0.5$, so each day maps to an L -bit pattern.

Archetypes. The reference is an archetypal analysis with $k = 5$ computed independently on the same series (A1–A5, Section IV-E). We compare (i) day labels, the dominant archetype against the Softmax state, by normalised mutual information (NMI, 0 for independent and 1 for identical partitions) and purity (share of days whose state’s majority archetype is their own); and (ii) profiles, the cosine similarity between each archetype’s taxon weights and the visible mean $\mu(\mathbf{e}_j)$ produced by each hidden unit alone.

Reconstruction. On each run’s own validation days, one taxon at a time is set to zero and the day is reconstructed by one up–down pass, reading the NB mean; we report the absolute (L1) error on the masked taxon, the mean L1 error and the cosine similarity on the other taxa, with 20 repetitions per taxon and day, averaged over the 10 seeds. ReLU families (divergent seeds) and binary families (0/1 data) are excluded.

Imputation. On the 160 days with missing taxa, the missing entries are initialised at zero and imputed by Gibbs sampling with the observed taxa clamped ($5 + 3 \times \text{\#missing steps}$); the score is the count NLL of the observed taxa under 100 hidden samples, per observed taxon. Each family is evaluated at its best L .

TABLE 2. All bounded count families fit stably; NB with Bernoulli or Sigmoid read-out fits best. Final validation metric at $L = 6$, shuffled split, mean $\pm$ std over 10 seeds (count models: NLL per entry, comparable only between models on the same scaled data; binary models: negative PLL, a different scale). Lower is better.

Visible	Hidden	Val. metric
NB	Bernoulli	0.444 ± 0.015
NB	Sigmoid	0.443 ± 0.013
NB	Softmax	0.475 ± 0.016
NB	ReLU	0.607 ± 0.110
ZINB	Bernoulli	0.491 ± 0.016
ZINB	Sigmoid	0.473 ± 0.019
ZINB	Softmax	0.498 ± 0.016
ZINB	ReLU	0.633 ± 0.407
Binary (median)	Bernoulli	0.475 ± 0.006
Binary (zero)	Bernoulli	0.287 ± 0.003

IV. RESULTS

A. Experimental Overview

Tab. 2 lists the ten families and their validation fit at $L = 6$ on the shuffled split. All count families with bounded hidden units train in 10/10 seeds. NB-Bernoulli and NB-Sigmoid share the same energy and training and tie, as expected; Softmax costs 0.03 in NLL for its one-state-per-day constraint; ZINB families are behind NB at the same budget, a gap we trace to non-convergence in Section IV-G.

B. Training Stability on Raw Counts

Three failures appeared in sequence. First, the NB mean $\exp(\eta)$ overflowed in float32 (NB, $L = 10$, epoch 161), producing NaN; clamping $\eta \leq 10$ removed it without touching any real value. Second, with CD-1 about 10% of seeds diverged at $L \geq 5$ at every learning rate tried. The dispersion trajectories grow identically at every L , which rules out θ drift; gradient clipping bounded the size of the updates but not their direction and was reverted (worst $L = 5$ seed improved from 0.87 to 0.67, but $L = 6$ failures rose from 1 to 2). We attribute the divergence to mode-locked chains: a one-step chain started from a bloom day stays in the bloom mode, and vice versa, so the negative phase is biased. With PCD the chains move between modes across updates; NB at $L = 10$, which before PCD diverged (NaN NLL), and every bounded family then trained in 10/10 seeds up to $L = 10$ (Fig. 2). Third, ReLU hidden units remained unstable: the $h^2/2$ restoring term of the rectified Gaussian energy assumes standardised inputs, while raw counts up to 444 drive the hidden input to ~ 44 per taxon; the $[0, 5]$ clamp stabilises single runs, but 6 of 10 seeds still diverge, and we discard ReLU. Sigmoid read-outs had 0 divergences in 50 runs.

C. Choice of L

Fig. 3 shows the validation NLL of NB-Bernoulli against L on both splits. On the chronological split the curve has an elbow at $L = 6$: the step $5 \rightarrow 6$ gains 1.6 seed standard deviations, the step $6 \rightarrow 7$ only 0.2. On the shuffled split the

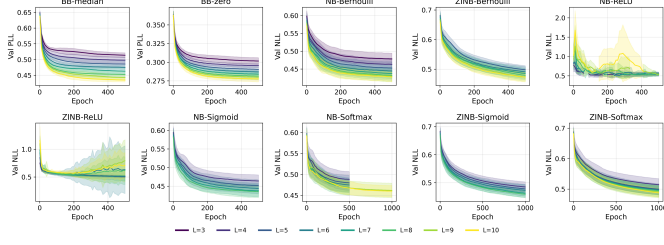


Fig. 2. With PCD, every bounded family trains smoothly up to $L = 10$. Validation metric against epoch for each family (panel) and L (colour), shuffled split, mean $\pm 1\sigma$ over 10 seeds; NLL for count models, negative PLL for binary ones.

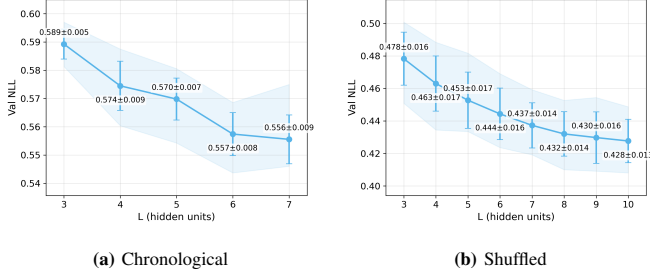


Fig. 3. $L = 6$ is the last step that beats seed noise on the chronological split; on the shuffled split no step does. Final validation NLL of NB-Bernoulli against L , mean $\pm$ std (bars) and min-max (band) over 10 seeds.

NLL keeps decreasing up to $L = 10$, but every step is below one standard deviation (≈ 0.015 ; e.g. $5 \rightarrow 6$: 0.0085, $6 \rightarrow 7$: 0.0070). We fix $L = 6$ for all families; Section IV-D shows that this is also the number of states a Softmax layer chooses on its own.

D. How Many Community States

A Softmax hidden layer assigns each day to one unit, so the number of units it keeps active measures how many states the data support. Offered $L = 4, \dots, 10$ units, the NB-Softmax RBM follows $y = L$ up to $L \approx 5$ and then saturates (Fig. 4): it uses 5.8 ± 0.8 units at $L = 7$ and 6.2 ± 0.6 at $L = 10$, never fewer than 5 nor more than 7 per seed from $L = 7$ on. The ZINB-Softmax RBM, with a different visible likelihood, gives the same plateau (5.5 ± 0.5 at $L = 7$, 6.5 ± 1.0 at $L = 10$), within one standard deviation of NB at every L . The plateau is thus a property of the data, not of the visible model.

E. Comparison with Archetypes

The reference. The archetypal analysis, provided by the research group supervising the course (unpublished), describes each day as a mixture of five archetypes (Fig. 5a). Each is led by one taxon: A1 by *Dinobryon* (80% of its weight), A2 by *Aulacoseira* (96%), A3 by cryptophytes (44%), A4 by colonial green algae with cyanobacteria, A5 by centric diatoms (54%). Their top-5 taxa (Fig. 5b) share a common background: cryptophytes are in all five sets, rotifers and nauplii in three; the taxa specific to one archetype are what separates them.

Softmax states against archetypes. Labelling each day by its dominant archetype and by its Softmax state gives two partitions of the same 1906 days (Fig. 6). Three archetypes

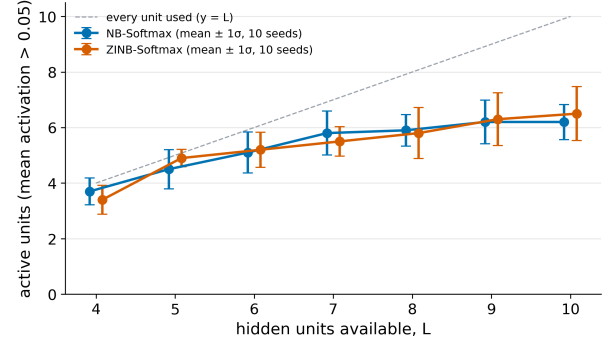


Fig. 4. Offered up to 10 one-hot states, the RBM uses about six, for both NB and ZINB visible units. Number of Softmax units with mean activation > 0.05 against L , mean $\pm$ std over 10 seeds, shuffled split.

TABLE 3. Four of five archetypes have a dedicated hidden unit; the cryptophyte archetype shares its unit with the centric diatom. Best-matching unit and cosine similarity between archetype profile and unit visible profile, NB-Sigmoid, $L = 6$, shuffled split, best seed.

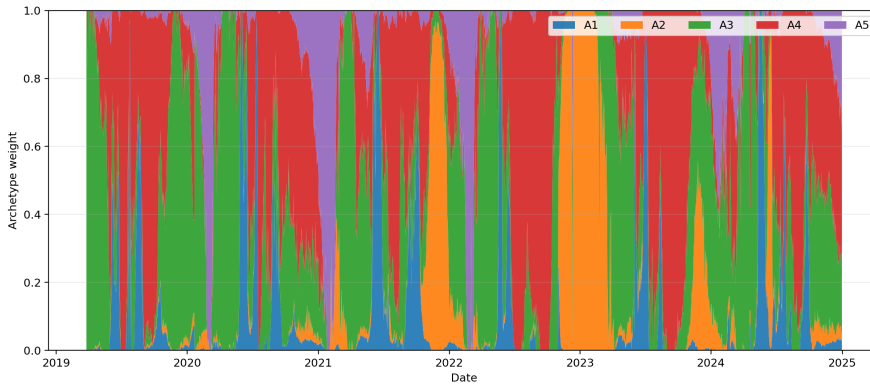
	Community	Unit	Cosine
A1	<i>Dinobryon</i> (chrysophyte)	H_3 (H_5)	0.73 (0.68)
A2	<i>Aulacoseira</i> (diatom)	H_0	0.54
A3	Cryptophytes	H_4	0.58
A4	Green algae + cyanobacteria	H_1	0.78
A5	Centric diatom	H_4	0.62

are captured by one state each: A2 by h_2 (94% of its days), A5 by h_4 (99%) and A1 by h_5 (96%). The states are not exclusive, however: h_5 also takes 47% of the A4 days and h_4 41% of the A3 days, while A3 and A4 are each split over two states. Overall NMI is 0.45 and purity 71%, a clear agreement but not an identity. The figure uses the best seed at $L = 6$, the L of the whole study, which leaves one unit without days; this is typical at $L = 6$ (5.1 ± 0.7 units used), while the six-state plateau of Section IV-D starts at $L = 7$. The choice does not move the result: averaged over the 10 seeds, NMI is 0.44 ± 0.03 at $L = 6$, 0.45 ± 0.03 at $L = 7$ and 0.46 ± 0.02 at $L = 10$, and purity stays between 70% and 72%.

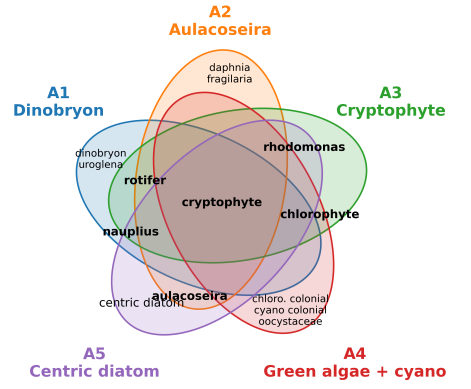
Final-model units against archetypes. In the NB RBM with Sigmoid read-out ($L = 6$, best seed) a day is a combination of units rather than a single state. All six units are used (mean activations between 0.24 and 0.81), 34 of the 64 possible on/off patterns occur, 10 patterns cover 80% of the days and 15 cover 90% (Fig. 7). Tab. 3 matches each archetype with the unit whose visible profile is closest. Four of five communities are recovered: summer green algae with cyanobacteria (A4, cosine 0.78), the chrysophyte *Dinobryon* (A1, 0.73, matched by two units, H_3 and H_5 at 0.68), the winter centric diatom (A5, 0.62) and *Aulacoseira* (A2, 0.54). The cryptophyte archetype A3 is blurred: its best unit (0.58) is the one already matched to A5. This is specific to the final model: NB-Bernoulli on the same split isolates A3 with cosine 0.92.

F. Reconstruction and Imputation

Tab. 4 compares the six count families on masked-taxon reconstruction. NB with Sigmoid read-out is first or tied on



(a) Archetype weights per day



(b) Top-5 taxa per archetype

Fig. 5. The reference archetypes alternate with the seasons and are each led by one taxon over a shared background of cryptophytes, rotifers and nauplii. (a) Weights of the $k = 5$ archetypes per day (sum to 1). (b) Five-set Venn diagram of the five largest-weight taxa of each archetype; bold labels are taxa shared by two or more archetypes.

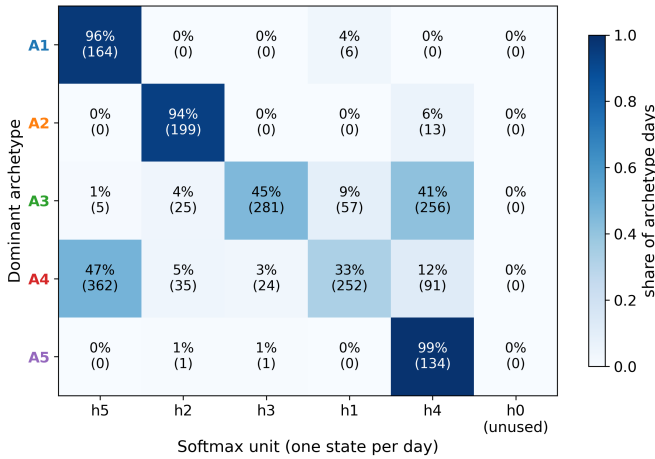


Fig. 6. Softmax states and dominant archetypes agree on 71% of the days; A1, A2 and A5 each map to one state, A3 and A4 are split and partly merged. NMI 0.45. Share of each archetype's days assigned to each NB-Softmax unit ($L = 6$, best seed, shuffled split; counts in brackets; h_0 wins no day in this seed). Units are ordered by the archetype whose days they capture the largest share of.

every metric, but the four non-Softmax families lie within about one seed standard deviation of each other; the clear effect is the Softmax constraint, which raises the error on the unmasked taxa by about 30% (0.68 vs 0.51) and lowers the cosine from 0.87 to 0.76. For the final model the error on the unmasked taxa is ≈ 0.53 and the cosine ≈ 0.87 whichever taxon is masked, while the masked-taxon error is concentrated on bloom formers: *Aulacoseira* ($L1 \approx 19$), *Dinobryon* (≈ 6) and cryptophytes (≈ 4).

Fig. 8 shows what these errors mean on one validation day drawn at random (2022-11-08, during the autumn *Aulacoseira* season). Most taxa are reconstructed within a fraction of a count (median absolute error 0.03–0.05), but *Aulacoseira* is overestimated, 36 true against 172 (Sigmoid read-out) and 188 (Bernoulli, averaged over 500 hidden samples). This single taxon accounts for most of the day's mean L1 error: 1.83 and 1.99 over the 83 taxa, 0.19 and 0.17 without it.

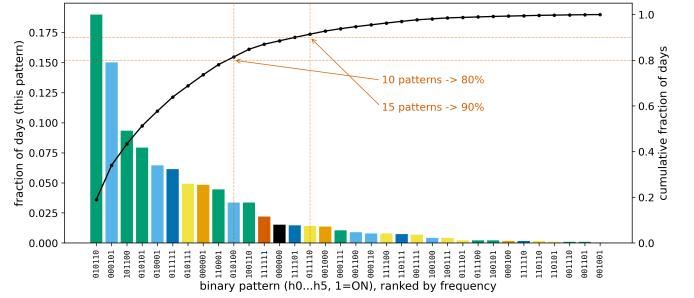


Fig. 7. In the final model a day is a combination of units: 34 of 64 patterns occur and 10 cover 80% of the days. Frequency of each 6-bit pattern, a unit being on if $p(h_j=1 | \mathbf{v}) \geq 0.5$ (bars, coloured by number of active units), and cumulative coverage (line), NB-Sigmoid $L = 6$, best seed, shuffled split.

TABLE 4. Masked-taxon reconstruction: NB-Sigmoid is first or tied; the one-hot Softmax layer costs about 30% on the unmasked taxa. Mean absolute error (scaled counts) and cosine similarity, $L = 6$, mean $\pm$ std over 10 seeds, each scored on its own validation days.

Family	Masked L1 ↓	Other L1 ↓	Cosine ↑
NB-Sigmoid	0.604 $\pm$ 0.022	0.509 $\pm$ 0.017	0.869 $\pm$ 0.008
ZINB-Sigmoid	0.604 $\pm$ 0.027	0.514 $\pm$ 0.027	0.869 $\pm$ 0.021
NB	0.609 $\pm$ 0.022	0.516 $\pm$ 0.012	0.860 $\pm$ 0.010
ZINB	0.613 $\pm$ 0.023	0.534 $\pm$ 0.025	0.846 $\pm$ 0.014
ZINB-Softmax	0.683 $\pm$ 0.031	0.667 $\pm$ 0.037	0.760 $\pm$ 0.022
NB-Softmax	0.700 $\pm$ 0.052	0.679 $\pm$ 0.055	0.761 $\pm$ 0.042

On the 160 days with structurally missing taxa (Fig. 9), NB-Sigmoid ($L = 7$) has the lowest NLL in the 31- and 54-missing patterns (0.497 and 0.306) and NB-Bernoulli ($L = 8$) in the 3-missing pattern (0.367). The day-to-day spread (± 0.13 –0.19) is, however, larger than every between-family difference, so no family is clearly better at imputation; ZINB-Sigmoid ties NB-Sigmoid on the 54-missing pattern (0.307).

G. Failure Modes: Zero Inflation

ZINB should suit plankton, where a taxon is either absent or present and possibly blooming, but in our budget it does not beat NB. At 500 epochs its NLL, θ and π are all still moving, and $\pi \approx 0.06$ is small and still decreasing, so the model is still

(iv) The chronological validation covers a single year, so the selected L mixes model capacity with the 2024 shift.

The most informative next experiment is to train ZINB to convergence, with longer runs, better mixing (mean-field or fast-weight PCD) and a hidden-dependent π : if it separated presence from abundance, its archetype match for bloom formers such as *Aulacoseira* would improve and the cryptophyte would get its own unit; if not, zero inflation adds nothing at this resolution. Repeating the state analysis with a count-decoder variational autoencoder would confirm whether the six-state plateau is a property of the data or of the RBM.

DATA AND CODE AVAILABILITY

The plankton time series is the Lake Greifen imaging dataset described in [1]. Code, configurations and every script that produced the figures and tables are available on GitHub; trained weights are not distributed and are regenerated by the training script.

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